Points Left on the Table

An open expected-points value layer and an outcome-validated decision-regret metric for NBA tracking data

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Abstract

The box score records what a basketball player *did*; it is structurally blind to the quality of the *decisions* that produced it. We build, on the only frozen public season of NBA optical tracking (2015–16), an open per-action expected-points value layer (an EPV "eval bar") and use it to define PLOT ("Points Left On the Table"): a per-decision *regret* — the gap between the expected points of the best genuinely-available option and the option the player chose, scored by the model's expected value rather than the realized outcome, so it measures decision quality and not luck. We validate the pipeline through a sequence of pre-registered gates: the value layer is calibrated under leave-one-game-out cross-validation (log-loss 0.82 vs. a 1.09 constant baseline; calibration-in-the-large −0.011; expected calibration error 0.063 points); the one-step counterfactual inputs are calibrated on held-out real continuations; and the resulting player metric is split-half reliable (Spearman–Brown 0.68), survives controls for shot location and finishing skill, and is largely orthogonal to the box score (joint R^2 with true-shooting and usage ≈ 0.07). The central question — does "points left" predict anything *real*, or only the model against itself? — is answered with two outcome-validity tests on 208 games. *Declining your own open look* costs real points: net of observables, team, and shot-clock pressure, passing up an open look costs ≈ 0.11 points on an average look and ≈ 0.19 on a high-value one; declining a *bad* look is free; the cost grows monotonically with look value; and it survives removing turnover exposure and action-layer mis-typing. The obvious selection objection cuts the wrong way: decliners realize *fewer* points than matched takers, not more, so on average the pass was not concealing a better option the tracking missed. A second, counterintuitive finding comes from the symmetric test — *shooting over a wide-open teammate* — which does *not* validate: the teammate-kick counterfactual is flat in its modeled value and shooting over the open man realizes *more* points, not fewer. So PLOT is outcome-valid precisely for the decision whose counterfactual is directly observable (your own open shot), while the conventional "pass to the open man" read is *not* recoverable from public tracking — whether because the open man is open precisely *because* he is not a threat, or because the teammate counterfactual is simply too crude to score, the read is not one this data can credit. Both validations are at the *decision* level; an out-of-sample cross-fit finds only *weak, fragile* support for clean attribution to individual *players* (pooled within-role $r \approx 0.11$, heavily attenuated from an in-sample 0.33, with one of two folds individually null), so we keep the validated claim at the decision level and report player-level attribution as modest at best. A variance decomposition shows the between-player signal is majority within-role decision quality (52%) rather than role-of-touch (29%), and the within-role component is itself reliable and robust to the strength of the role control. We release the value layer, the metric, a chess.com-style possession-review demo, and all gate code. We are

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explicit about scope: PLOT measures genuinely-left value *conditional on observables*, not proven-causal "decision IQ," and the validated claim is narrow, modest in magnitude, and honest about selection on unobservables.

1 Introduction

A great pass that leads to a missed layup and a forced heave that happens to go in are recorded by the box score in exactly the wrong order. Counting statistics — and even the efficiency metrics built on them — evaluate *outcomes*, which conflate decision quality with finishing skill and luck. Yet coaches and analysts talk constantly about *decisions*: the open shot a player passed up, the kick to the corner he didn't make, the contested two he took over an open teammate. Player and ball tracking make these decisions observable for the first time, but the dominant tracking-era value model — expected possession value (EPV) [1, 2] — stops just short of scoring them. Cervone et al. introduced EPV and, alongside it, "shot satisfaction" (the value a shot adds relative to the possession's EPV): an essentially shot-centric, opposite-signed cousin of regret. A possession-level *regret* — best genuinely-available option minus the chosen one — on *public* data has remained a gap.

This paper closes part of that gap and is honest about the part it cannot. We make five contributions.

1. **An open value layer.** A reproducible per-action expected-points model for NBA possessions — a "VAEP-for-basketball" [5] — trained from public tracking, with a permutation-invariant set encoder over the ten players feeding a causal recurrent outcome head (§4). It is calibrated on held-out games and its per-action ΔEPV decomposition telescopes exactly to realized possession points.

2. **The PLOT regret metric.** A per-decision regret valued against the best genuinely-available alternative and scored by model expected value, aggregated to a per-player "points left on the table" (§5). We show it is a stable, distinct skill signal, not a relabeling of shot location, finishing, or the box score (§6).

3. **Outcome validity at the decision level (the keystone).** Using realized possession outcomes as ground truth (§7.2), declining your own open look costs real points in a dose-responsive way that survives every robustness cut we could devise. The natural selection objection cuts the *wrong* way: decliners' realized outcomes are *worse*, not better, than matched takers', so on average these passes were not justified by an unobserved superior option. These validations are at the decision level; an out-of-sample cross-fit gives only weak, fragile support for clean attribution to individual *players* — pooled within-role $r \approx 0.11$, one of two folds null, heavily attenuated from an in-sample 0.33 (§7.4).

4. **The open-man myth (a finding in its own right).** The symmetric test — *shooting over a wide-open teammate* — does *not* validate: the teammate-kick counterfactual is flat in its modeled value, and shooting over the open man realizes *more* points, not fewer (§7.3). Either an open man is often open precisely *because* he is not a threat, or the teammate counterfactual is simply too crude to score from tracking (we cannot fully separate the two); either way the conventional "pass to the open man" read is not recoverable from public tracking. This is a result about the limits of what such tracking can credit, not merely a limit on PLOT.

5. **Artifacts.** The value layer, the metric, a chess.com-style possession-review web demo (§8), and the full gate-by-gate reproduction code are released openly.

Throughout we treat the metric with deliberate skepticism. Every stage is a go/no-go gate, and the most important gate (outcome validity) is the first one that is *not* the model judged against itself. We report a negative result (the teammate-kick test) as prominently as the positive one, because it is what scopes the claim. And we are clear that the headline magnitude is modest — declining a great open look costs about a fifth of a point — which is the honest size of a real, previously-unmeasured effect, not a dramatic one.

2 Related work

Expected possession value. EPV [2] models a possession as a multiresolution stochastic process and reads off the expected points conditional on the full spatiotemporal state. It is the conceptual parent of our value layer. Our contributions relative to it are (i) an open, lean reimplementation on public data, and (ii) building the possession-level regret that EPV named but did not develop. Sicilia et al. [12] (DeepHoops) similarly evaluate micro-actions with deep spatiotemporal features; we adopt that spirit with a smaller, calibration-first model and an explicit decision metric on top.

Action values in invasion sports. VAEP [5] values every soccer action as the change in scoring/conceding probability it induces; our per-action ΔEPV layer is the direct basketball analog, with possession EPV playing the role of VAEP's state value.

Shot quality and shooting strategy. Shot-quality models such as qSQ/qSI [3] estimate the value of a shot from its spatial context; we use a deliberately lean "xPOINTS" shot model of the same family as the open-shot counterfactual. Sandholtz and Bornn [10], Sandholtz et al. [11] study shooting strategy and spatial allocative efficiency as decision problems; our regret is a complementary, per-decision, outcome-validated quantity. Spatial-structure models of offense and defense [6, 9] inform our state features.

Box-score baselines. Efficiency and usage summaries [7] are the incumbents we test PLOT against for incremental information; the point of PLOT is precisely to measure something they cannot.

Architecture. The value layer uses a Deep Sets [13] encoder for permutation invariance over players and a GRU [4] for the causal temporal head.

3 Data and possession pipeline

We use public 2015–16 NBA SportVU optical tracking: 25 fps (x, y) for ten players plus the 3D ball, distributed through community archives [8] after the league's movement endpoint was closed, merged with play-by-play (PBP) on `eventId == EVENTNUM` ($\approx$98.5% match). 2015–16 is the last season with publicly available league tracking; coverage runs 2015-10-27 through 2016-01-23 ($\approx$ half a season, $\approx$636 games). **We do not re-host the raw data**: the repository ships a download script, derived features, and model outputs only, with every transformation documented.

Pipeline. Each game is parsed to an internal schema and downsampled to 10 fps parquet (the model resolution; 25 fps is retained only for demo clips). Possessions are segmented from PBP event types plus the controlling team, terminating on a made field goal (with any and-one free throws), defensive rebound, turnover, or period end; offensive rebounds *continue* the possession. Realized possession points are computed *description-based* from the authoritative scorer team id, never from the orientation-inconsistent and occasionally non-monotonic PBP `SCORE` column. Segmentation was validated by exact points reconciliation on every local tracking game and on a 205-game PBP-only sweep (a single residual traced to a known SCORE-column glitch).

Orientation and quarantine. Coordinates are canonicalized so offense always attacks the same basket. Because the raw feed carries no possession orientation, we infer it from spatial evidence and *quarantine* games where the inference is not corroborated (single-confident-cell, conflicting, or high-backcourt-fraction games), trading coverage for correctness. The headline corpus is the **208 clean games** that pass all quality control; the value layer's calibration gate (G1) is reported on a 10-game development subset for which both the baseline and sequence models were trained under identical leave-one-game-out (LOGO) folds. We note one caveat the trade-off carries: the retained games are not guaranteed missing-at-random — if particular arenas or camera installations produced systematically noisier tracking and were dropped, the clean corpus may be mildly non-representative of the league (§9).

Nested corpora. The analysis uses several samples; Table 1 names them and the gate each serves, so the reader need not reverse-engineer which N applies where. Development subsets validate the machinery; the 208-game clean corpus carries the headline results.

Table 1: **Nested corpora** and the gate each serves.

Sample	Size	Used for
Available half-season	≈636 games	full public 2015–16 coverage (Oct 27 – Jan 23)
Development subset	10 games (LOGO)	value-layer + counterfactual calibration (G1, G2)
Development corpus	42 games	validity method checks (G5)
Headline corpus	**208 clean games**	stability, distinctness, decomposition, validity (G3–G6)
Player panel	374–387 players	≥30 decisions; 374 matched to season box stats (G4)

4 The value layer: an open expected-points model

The value layer predicts, at every frame of a possession, the distribution over the points the possession will ultimately yield, $P(\text{points} = k)$ for $k \in \{0, 1, 2, 3\}$, and reads off the **eval bar**

$$\text{EPV}_t = \sum_{k\in\{0,1,2,3\}} k \cdot P(\text{points} = k \mid \text{state}_t). \tag{1}$$

We build two models with the *identical* target, weighting, calibration code, and quarantine, so the upgrade is an apples-to-apples comparison.

Baseline (LightGBM). A gradient-boosted multiclass head over coarse per-frame features (canonical ball geometry, court region, game/shot clock, score margin, period, paint counts, ball-handler-to-nearest-defender pressure). It is deliberately decision-insensitive — a location/context prior — and exists to validate the calibration plumbing.

Sequence model. A permutation-invariant **Deep Sets** [13] encoder over the ten players (offense/defense-masked mean+max pooling; per-player features are canonical position, team and ball-handler flags, and distance to the ball), concatenated with a ball token and context vector, fed to a **causal (unidirectional) GRU** [4] (hidden size 128, one layer) and a per-frame multiclass head. Causality is a tested invariant: editing frame t leaves all earlier frames' predictions bit-identical, so the eval bar at time t never peeks ahead. Training is on Apple MPS in minutes per fold.

Calibration (Gate G1). Both models are evaluated out-of-fold on held-out *games* (Table 2). The sequence model is well calibrated (calibration-in-the-large -0.011, moment slope 0.955 with 95% CI $[0.92, 0.99]$, expected calibration error 0.063 points) and improves log-loss, Brier, and the

Table 2: **Gate G1 — value-layer calibration**, leave-one-game-out on 10 development games ($\approx$3.1–3.2×10^5 held-out frames). Lower is better for log-loss/Brier/RPS; calibration-in-the-large (CITL) and ECE near 0; slope near 1. The sequence model is the eval bar used downstream.

Model	log-loss	Brier	RPS	CITL	ECE (pts)	slope
Constant-rate prior	1.086	0.615	0.597	—	—	—
Baseline (LightGBM)	1.014	0.569	0.545	+0.006	0.054	1.15
Sequence (Deep Sets + GRU)	**0.819**	**0.461**	**0.428**	−0.011	0.063	0.955

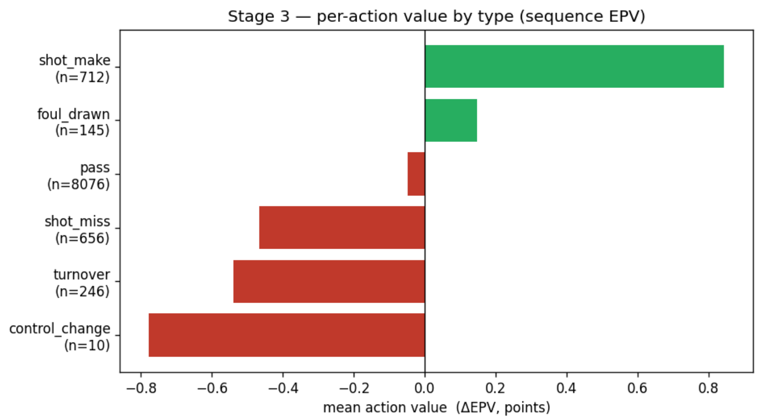

Figure 1: Per-action ΔEPV by action type. The decomposition telescopes exactly to realized possession points; signs and magnitudes are face-valid.

ordinal ranked probability score by $\approx$17–21% over the baseline, and far over a constant-rate prior. We report the calibration slope as a diagnostic, not a gate; the gate is (beats-constant-baseline) $\wedge$ (calibration-in-the-large) $\wedge$ (ECE). **G1 passes.**

Per-action decomposition. Following VAEP [5], each on-ball action is valued as value(a) = EPV(end) − EPV(start), grounded in the ball-handler trace, with the terminal action pinned to realized points. By construction the action values telescope to (realized points − initial EPV) over a possession: across 1,774 possessions and 9,850 actions the maximum absolute telescoping residual is exactly 0.0. The action-type means are sensible (Figure 1): made shot +0.84, foul drawn +0.15, pass −0.05 (mean $\approx$ 0, as a pass should be on average), missed shot −0.47, turnover −0.54.

5 The PLOT regret metric

Decision points and the available set. We score decisions at action boundaries. The narrowest, best-identified decision — and the v1 metric — is the *open pass-up*: a ball-handler with no defender within 4 ft who passes rather than shoots. The *best available* option is that open shot, valued by an xPoints model; the *chosen* value is the post-pass EPV read from the value layer. Regret is

$$\text{regret} \;=\; \underbrace{\text{xPoints(open shot here)}}_{\text{best available}} \;-\; \underbrace{\text{EPV(chosen action)}}_{\text{chosen, by model EV}}, \tag{2}$$

reported both clipped, $\max(0,\cdot)$ ("points left"), and signed (a diagnostic). Crucially the chosen action is scored by its *model expected value*, not its realized outcome: PLOT measures the decision, not whether it happened to work.

Table 3: **Gate G2 — counterfactual inputs**, calibrated against held-out realized outcomes.

Input	n	CITL	ECE (pts)	slope	beats baseline
xPoints (open shot, when taken)	1,233	+0.002	0.077	0.842	yes (log-loss 0.668<0.689)
post-pass EPV (when passed)	8,076	−0.014	0.066	0.915	—

The xPoints counterfactual model. xPoints = P(make | distance, openness, shot type) × points is a regularized logistic model fit on observed shots — well-identified because it is trained on real shots from similar states. It uses a population make model with *no* shooter identity, so regret cannot smuggle in finishing skill (verified in §6).

Gate G2 (one-step counterfactual calibration). Regret is only meaningful if its two inputs are calibrated on held-out *real continuations*. They are (Table 3): when the open shot *was* taken, xPoints matches realized points (calibration-in-the-large +0.002, ECE 0.077, and it beats a make-rate baseline on log-loss); when a pass *was* made, post-pass EPV matches realized outcomes (calibration-in-the-large −0.014, slope 0.915, ECE 0.066). **G2 passes**; regret is not a calibration artifact.

We later (§7) broaden the scored set to a *credible set* that adds shot-selection decisions (shooting over a wide-open, reachable, completion-discounted teammate), and analyze both the metric's composition and its outcome validity on that set.

6 Is it a skill? Stability, distinctness, validity

Gate G3 (stable and not finishing). On 208 games (124,953 open pass-up decisions, 380 players with ≥ 30), PLOT is split-half reliable: splitting the season's games odd/even and correlating per-player means gives Pearson 0.51 ($p \approx 0$), Spearman–Brown full-data reliability 0.68 (Figure 2). A nested OLS shows player identity explains regret *beyond* shot location ($F(380, 124568) = 4.96$, $p \approx 0$), and per-player regret is uncorrelated-to-mildly-negative with finishing skill (Pearson −0.17), so it is not a relabeled finishing metric. **G3 passes.**

Gate G4 (incremental over the box score). On 374 players matched to season box statistics, PLOT is essentially unrelated to true-shooting ($r = +0.09$, n.s.) and only weakly related to usage (−0.24), points per game (−0.21), and field-goal attempts (−0.24). A joint regression on true-shooting and usage explains just $R^2 = 0.075$ of PLOT variance — >90% is *not* in the box score (Figure 3). **G4 passes** (incremental ∧ reliable).

The leaderboard, and an honest caveat. Sorted by raw PLOT, the top is rim-running bigs and role players (Miles Plumlee, Montrezl Harrell, Kosta Koufos, Willie Cauley-Stein, Hassan Whiteside) and the bottom is high-usage lead guards (John Wall, Russell Westbrook, Bradley Beal, Kevin Durant). This is face-valid — bigs defer open looks; guards take them — but it is exactly the validity worry: is a big kicking out an open 18-footer a *bad decision* or correct *deference to role*? G4 establishes that PLOT is new information; it does not by itself establish that the information is decision *quality*. Sections 7.1–7.2 address this directly. One caution on the names: split-half reliability is moderate (Spearman–Brown 0.68 for raw PLOT, 0.54 for the within-role residual; §7.1) — high enough to establish that a real between-player signal *exists*, but not high enough to pin any single player's value precisely. Individual estimates carry meaningful sampling noise, so this leaderboard should be read as evidence that the signal is real, not as an exact ranking of these specific players.

Gate G5 (validity conditional on observables). Three checks (demonstrated on the 42-game development corpus; these are method properties that transfer): (a) the regret inputs are cali-

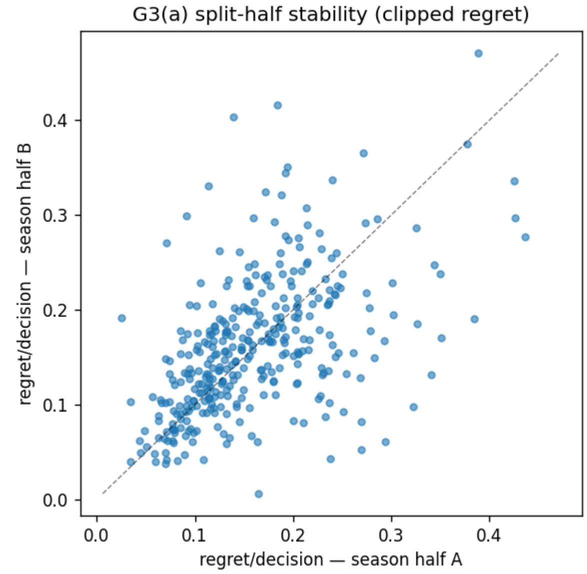

Figure 2: G3: split-half reliability of per-player PLOT across odd/even game halves.

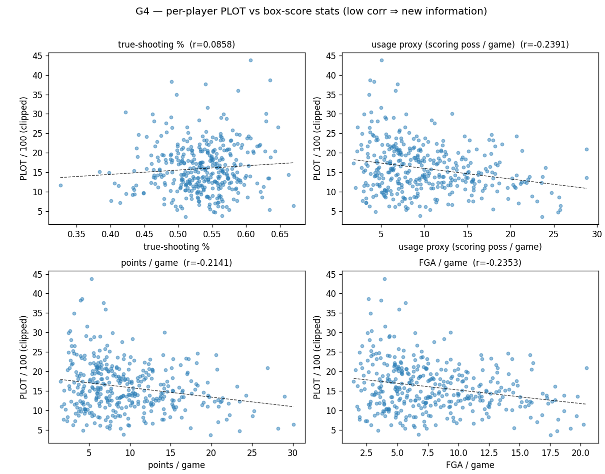

Figure 3: G4: PLOT vs. box-score efficiency/usage — weak to null, i.e. incremental information.

brated within the open-decision subspace, not just on average (post-pass EPV ECE 0.055, XPOINTS make ECE 0.040; near-rim XPOINTS is if anything biased *low*, which would *under*-credit bigs' passed-up rim looks — the opposite of the feared bias); (b) selection onto the decision is, as expected, strong (propensity AUC 0.82) but driven by *modeled* features (distance and openness — XPOINTS's own inputs), so the metric conditions on the selectors; (c) the per-player ordering is robust to finishing skill — a shooter-aware XPOINTS barely reorders the leaderboard (Spearman 0.93, top-30 overlap 0.80). The residual threat is selection on *unobservables* (shot difficulty beyond location and openness), which is irreducible without a counterfactual and is our principal standing caveat.

7 What does it measure, and is it real? (Gate G6)

Everything through G5 is *internal*: regret is the model compared against itself or its own construction. G6 asks the two questions that decide whether the paper says something true and useful: *what* does the between-player signal measure (role-of-touch vs. within-role decision quality), and does it predict anything *real* (outcome validity)?

7.1 Role-of-touch vs. within-role decision quality

Every variant of the metric puts bigs near the top, raising the worry that PLOT measures *where a player gets the ball* rather than *how well he decides*. We put a number on it. On the credible set (153,098 decisions, 387 players ≥ 30), we fit a **context model** regret $\sim$ touch context (court location, openness, rim distance, three-point indicator, decision kind) with *no* player identity, out-of-fold by game. The out-of-fold prediction is the **role** component (the regret anyone faces in that spot); the residual is the **within-role** component (how much this player beats or trails the situation's expectation). We then decompose the between-player variance via the exact identity $\mathrm{Var}(\mathrm{raw}) = \mathrm{Var}(\mathrm{role}) + \mathrm{Var}(\mathrm{resid}) + 2\,\mathrm{Cov}$.

Table 4: **G6.1 — role vs. within-role decomposition** of between-player variance (208 games, credible set). The within-role component is the majority of the variance and is reliable; reliability is stable as the role control strengthens (right), i.e. not leakage.

Variance share	value	Leakage sensitivity (role-model capacity)			
role share	0.290	capacity	weak	default	strong
within-role share	**0.525**	role share	0.149	0.290	0.325
cross (2 Cov)	0.186	within-role share	0.611	0.525	0.505
R^2 role $\rightarrow$ leaderboard	0.505	residual reliability (SB)	0.598	0.540	0.521
raw reliability (Spearman–Brown)	0.726				
within-role residual reliability (SB)	0.540 ($p \approx 0$)				

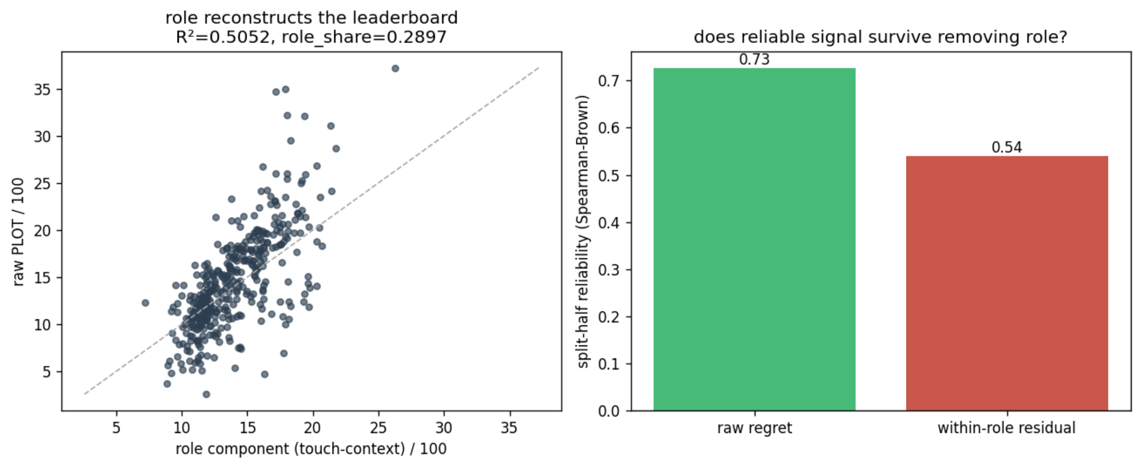

Figure 4: G6.1. Left: the role (touch-context) component reconstructs about half the raw leaderboard ($R^2 = 0.51$). Right: a reliable signal *survives* removing role — the within-role residual is split-half reliable, so there is real role-adjusted decision quality underneath the role-of-touch.

The result overturns the "it's just role" eyeball (Table 4, Figure 4): within-role decision quality is the *larger* component (within-role share 0.52 vs. role share 0.29), and the within-role residual is itself split-half reliable (Spearman–Brown 0.54, $p \approx 0$). A leakage check confirms this is genuine: as the role model is strengthened (weak $\rightarrow$ strong), its share rises then plateaus ($0.15 \rightarrow 0.29 \rightarrow 0.32$) while the residual's reliability stays *stable* ($0.60 \rightarrow 0.54 \rightarrow 0.52$) — if the residual were merely unabsorbed role, a stronger role model would eat it; it does not.

7.2 Outcome validity I — declining your own open look (the keystone)

This is the one test that is not model-vs-model. The cleanest decision the metric makes is the **open look**: a ball-handler with nobody within 4 ft either *takes* the shot or *declines* it (passes). We observe realized possession points for *both* choices, so the counterfactual "what taking would have yielded" comes from real shooters at matched situations, not from the model.

Design. On 208 games we assemble 79,877 open-offensive-player decisions: 11,365 *takers* — the real PBP shots, attributed to the player who actually shot (so the benchmark is the shot's own realized points) — and 68,512 *decliners* (open ball-handlers who passed). The benchmark S is the player's own open-shot xPoints (same model on both sides); the outcome R is realized possession points. We regress R on a `declined` indicator, S, their interaction, and observable

Table 5: **G6.2 keystone — realized points lost by declining an open look** (208 games; 79,877 decisions). Adjusted for observables, shot-clock proxy, and team fixed effects; 95% game-cluster bootstrap intervals. Positive = declining costs real points; the negative interaction means the cost grows with look value.

Quantity	estimate	95% CI	note
points lost, average look ($S \approx 1.10$)	0.112	[0.086, 0.137]	$p \approx 0$
points lost, high-value look ($S \approx 1.45$)	**0.193**	[0.141, 0.251]	$p \approx 0$
`declined` $\times S$ interaction	−0.231	[−0.351, −0.116]	cost grows with S
robustness: shot-ending only (no TOs), high-S	0.214	[0.153, 0.278]	survives
robustness: clean decliner pool (drop ∼28%), high-S	0.215	[0.169, 0.266]	survives

controls — the model's EPV at the decision, location, openness, *seconds into the possession* (a shot-clock-pressure proxy that also corrects for takers acting later than decliners), and period — plus team fixed effects, with a **game-cluster bootstrap** so inference respects within-game dependence. The validity signature is that the cost of declining is positive and *grows* with S.

Result (Table 5, Figure 5). It is. Net of observables, clock, and team, declining an open look costs 0.112 points $[0.086, 0.137]$ on an average look and 0.193 $[0.141, 0.251]$ on a high-value one ($S \approx 1.45$); the `declined` $\times S$ interaction is -0.231 $[-0.351, -0.116]$ (the cost grows with look value); and the *raw* cost runs from ≈ -0.005 for a bad look (declining it is free — correct, and cheap) to $\approx +0.42$ for a great one. **The gate passes at the decision level.**

Why it is not an artifact. (i) *Dose-response*: a pool-composition confound would be flat in S; this is sharply increasing. (ii) *Turnover exposure ruled out*: restricting to possessions that end in a field-goal attempt (both groups in the same structural position) leaves the high-value cost at 0.214 $[0.153, 0.278]$ — declining a good look yields a genuinely *worse shot*, not just turnover risk. (iii) *Not action-layer mis-typing*: dropping every decliner whose handler took a PBP shot that possession (19,466 rows, $\approx 28\%$) barely moves the estimate (0.133 mean / 0.215 high-value). (iv) *Against selection-on-unobservables*: if decliners passed because they saw a better play, their realized outcome should be *better*; it is *worse* — so on average these declines were not justified by an unobserved superior option (individual ones may be; the bound is on the average, per the G5 caveat).

Magnitude, honestly. A 42-game pilot put the high-value cost near 0.47; the well-powered 208-game estimate is ≈ 0.19. The pilot figure was small-sample-inflated. The *sign, dose-response, and every robustness cut hold*; the effect is real and modest, not large.

Magnitude in context. A fifth of a point per decision invites the question of season-level stakes, but the aggregate is easy to get badly wrong, and we decline to report a naive one. "Open" (no defender within 4 ft) is a loose, frequently-satisfied condition, so a ball-handler declines an open look many times per game — yet most of those declines are *correct* (low-value looks, where the cost is ≈ 0; the signed regret is negative on average, i.e. passing usually beats the contested-spot shot), and the cost concentrates on the minority of *high-value* declines. Multiplying the per-decision cost by the full decline frequency therefore overstates the stakes by more than an order of magnitude, and the raw, *unadjusted* bin costs additionally fold back in the selection the keystone regression removes. A defensible season- or player-level aggregate needs the per-player frequency of *high-value* open-look declines weighted by the adjusted cost — precisely the player-level quantity the cross-fit (§7.4) is built to estimate — so we report the validated effect at the decision level and defer the aggregate to that analysis.

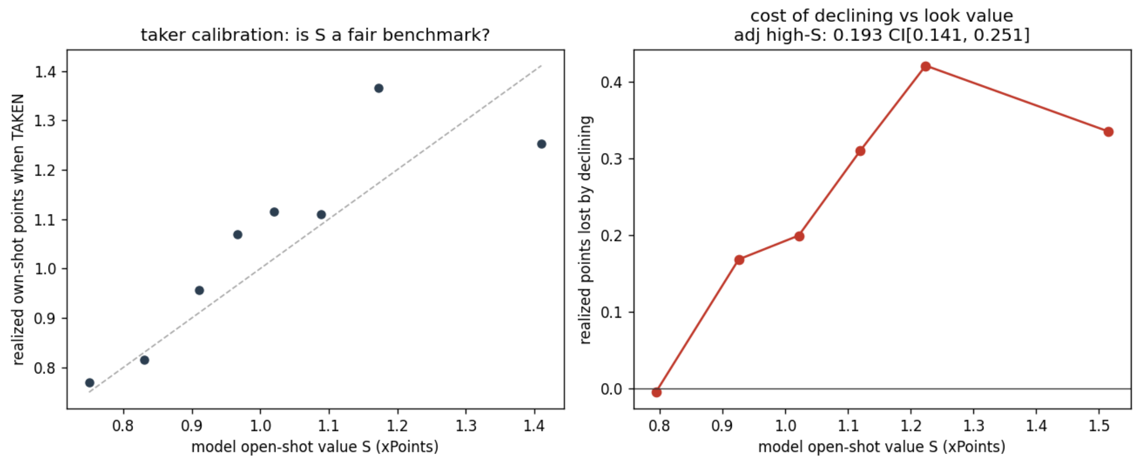

Figure 5: G6.2. Left: taker calibration — open looks of model value S realize $\approx S$ when actually taken, so S is a fair benchmark. Right: the realized cost of declining rises with look value — free for bad looks, costly for good ones — exactly the dose-response regret predicts.

Table 6: **G6.2b — shooting over a wide-open teammate does not validate** (208 games; 56,785 decisions). The teammate-kick counterfactual is flat in its modeled value, and shooting over the open man realizes *more* points, not fewer.

Quantity	estimate	95% CI	note
"points lost" by shooting over, average	-0.220	$[-0.256, -0.185]$	wrong sign
"points lost" by shooting over, high-S	-0.228	$[-0.272, -0.190]$	wrong sign
`declined` $\times$ S interaction	$+0.031$	$[-0.049, 0.122]$	no dose-response ($p = 0.46$)
gate		**FAIL** (correctly)	

7.3 Outcome validity II — the open-man myth (a defining negative)

We run the symmetric test for the other half of the credible set: *shooting over a wide-open teammate.* When a handler has a wide-open ($\geq$6 ft), frontcourt, reachable teammate, does shooting over him cost points versus kicking to him? The benchmark S is the kick's value ($0.80\times$ the best open teammate's xPoints); `declined` now means *shot over the open man*; same controls, team fixed effects, and bootstrap (56,785 decisions: 14,761 shot over, 42,024 kicked).

It does *not* validate, and the opposite is true (Table 6). The teammate-kick benchmark is *flat* in S (kick calibration ≈ 1.0 regardless of modeled value), the adjusted "cost" of shooting over is *negative* (-0.228 $[-0.272, -0.190]$: shooting over realizes *more* points), there is no dose-response (interaction $+0.031$, $p = 0.46$), and the gate correctly **fails**.

This negative is one of the most valuable results in the project. (i) *It explains why "score more decision types" fails.* An earlier variant that valued every decision against the best open teammate collapsed into a position sort precisely because this counterfactual is unreliable: an "open" man is often open *because* he is not a threat, so the kick does not deliver its modeled value. We had diagnosed that by face validity; outcome validity now confirms it. (ii) *It scopes the claim exactly.* PLOT is outcome-valid for the decision whose counterfactual is directly observable — declining your own open look, whose value is the shooter's own calibrated xPoints — and *not* for the "should've passed to the open man" read, contradicting conventional basketball wisdom. Realness comes from a clean foundation plus scale plus the one validated decision, not from breadth.

7.4 Outcome validity III — player-level cross-fit (de-circularized)

The decision-level keystone is non-circular by construction. The *player-level* read — "does PLOT identify *who* leaves points?" — is the prize, and the naive version is circular: correlating a player's model metric with their own realized shortfall $S - R$ reuses the model's S on both sides (mechanical) and measures both on the same games. An in-sample version of this correlation is positive (within-role residual $r = 0.33$, $n = 373$), but we treat it as exploratory only.

Method (the rigorous version). We remove both leaks with a cross-fit. Split the season's games odd/even. On one half compute the player's model metric (raw PLOT and the within-role residual, per 100 decisions). On the *other* half compute their realized points-left as a **matched-taker benchmark**: for a declined open look of value bin b, the realized cost is

$$\text{cost} \;=\; \underbrace{\overline{R}_b^{\,\text{takers}}}_{\substack{\text{mean realized points of real}\\ \text{takers in the same } S\text{-bin}}} \;-\; \underbrace{R}_{\substack{\text{decliner's}\\ \text{realized points}}}, \tag{3}$$

averaged per player. The benchmark is *empirical taker outcomes*, not the model's S (which now only defines the matching bins), so the two axes share no construction *and* come from disjoint game sets. We run both directions (metric-on-A/cost-on-B and the swap), pool, and require the pooled within-role correlation to be positive, significant, and sign-consistent across directions. A positive result is genuine, non-circular player-level outcome validity; a null is an honest "decision-level real, player-level attribution not clean."

Result — weak, positive, and fragile. Run over 207 of the 208 games (one game's source archive was corrupt; immaterial), the cross-fit gives a *modest* signal. Pooled across both directions ($n = 659$ player-halves), the within-role residual correlates with realized points-left at Pearson $r = 0.11$ ($p = 0.004$; raw PLOT $r = 0.10$, $p = 0.009$), and both directions share the positive sign, so the pre-registered gate passes. But the result is *fragile*: the two directions disagree sharply — model-on-half-B / cost-on-half-A gives $r = 0.19$ ($p = 0.0005$), while the reverse is individually *null* ($r = 0.05$, $p = 0.33$). Read honestly against the in-sample value, the player-level signal *shrinks* as the circularity and in-sample optimism are stripped out: 0.33 (in-sample, shared S, same games) $\rightarrow$ 0.11 (out-of-sample, empirical taker benchmark). We therefore report this as *modest, non-circular support* that the metric carries *some* player-level attribution — not a clean confirmation. The robust, outcome-validated claim remains at the *decision* level (§7.2); any single player's PLOT should be read with the wide uncertainty this and the moderate split-half reliability (§6) together imply.

8 The demo

The value layer and metric are exposed through a chess.com-style possession-review web app (Next.js): an animated court, an **eval bar** that rises and falls with EPV, and per-decision **badges** (Great/Good/Inaccuracy/Mistake/Blunder, thresholded on points left) at open pass-ups. All data are precomputed to static JSON — there is *no* model inference in the browser — so the demo is a thin, deployable view over the same artifacts the gates use. It makes the abstract metric legible: you watch a possession and see, frame by frame, where the points went. It is live at https://plot-nba.vercel.app.

9 Limitations

We state the ceiling plainly.

- **Scope.** Exactly one decision is outcome-validated — declining your own open look. Drives, and the "pass to the open man" read, are explicitly *not* validated (§7.3); the metric is a narrow, well-identified instrument, not a general "decision IQ."
- **Selection on unobservables.** Conditioning on observables (including the model's EPV at the decision) bounds selection on *observables*; it cannot rule out that decliners see something the tracking does not. The keystone's wrong-direction-for-selection argument bounds this on average, but does not eliminate it for individual decisions.
- **Role-of-touch.** About a third of the between-player signal is role-of-touch (§7.1); the honest object of study is the reliable within-role residual, which we isolate but which is harder to communicate than the raw leaderboard.
- **Magnitude.** The validated effect is modest (≈ 0.19 points on a great look). It is a real, previously-unmeasured quantity, not a dramatic one. We deliberately do not convert it into a season-level aggregate (§7.2): the honest one requires a player-level frequency the decision-level analysis does not supply.
- **Reliability of individual estimates.** Split-half reliability is moderate (Spearman–Brown 0.68 raw, 0.54 within-role), which supports the existence of a real between-player signal but means any single player's value is estimated with meaningful noise. The leaderboards should be read as evidence the signal exists, with uncertainty, not as exact rankings; we do not attach per-player intervals because the gate outputs do not currently include them.
- **Corpus representativeness.** The orientation quarantine retains $\approx$208 of $\approx$636 games. This is the right correctness trade-off, but the retained set is not guaranteed missing-at-random: if certain arenas, camera installs, or teams produced systematically worse tracking and were dropped, the clean corpus may be mildly non-representative of the league.
- **Data.** Public tracking covers half of one season; the 10 fps action layer can mis-type an offensive-rebounded shot as a pass (shown not to drive the keystone, §7.2).

10 Conclusion

Decision quality in basketball is measurable from public tracking, it is a stable and box-invisible player signal, and — for the one decision whose counterfactual is directly observable — it is *outcome-valid*: players systematically leave real points by declining good open looks, in a dose-responsive way that survives every robustness cut. The mirror-image read favored by basketball intuition — "pass to the open man" — is, by contrast, *not* recoverable from the same data: it fails the identical test, whether because the open man is open precisely because he is not a threat or because that counterfactual is too crude to score from tracking alone. That asymmetry is a result in its own right, and it disciplines the claim to exactly what the data support. The honest size of the effect is modest, the honest scope is narrow, and the honest object is the role-adjusted within-role signal — but it is real, and it is something the box score cannot see. We release the value layer, the metric, the gate code, and the demo so the result can be checked, extended, and argued with.

Reproducibility and data availability

All models, gates, and figures are reproducible from the released code; the raw tracking data are not re-hosted but are downloadable from the documented public archives [8] via the included script. Each gate (G1–G6) writes a machine-readable report (calibration tables, bootstrap intervals, leaderboards) and a figure; the per-player metric and within-role residual are emitted as columnar tables. The web demo serves precomputed JSON with no in-browser inference.

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